MIL-C-63513A (AR)
AMENDMENT 4
23 JANUARY 1989
SUPERSEDING
AMENDMENT 3
28 April 1988

MILITARY SPECIFICATION

CARTRIDGE, 105MM, TP-T, M490A1
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-63513A (AR), dated 5 September 1985, and is approved for use within the U. S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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4.4.2.2: Change "Length of liner incorrect (Flush to 1/4 inch above mouth of case)" to "Length of liner incorrect (Flush to 1/2 inch above mouth of case)".

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4.4.2.3: Critical Defect: Add:

"2	Plug missing	100%	3.2	Visual"

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4.4.2.4: Delete:

"201	Evidence of poor workmanship	1.65%	3.8	Visual"

And substitute:

"201	Evidence of poor workmanship	0.65%	3.8	Visual"

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4.4.2.7: Delete:

"104, Gap between cartridge case and obturator, max. 0.40%, 3.2, Gage" and replace with:
"104, Gap between cartridge case and obturator, max. 100% 3.2 Gage"

Add the following:

"105, Forward bourrelet diameter, max. (Note 4), 100% 3.1 Ring Gage"

* Add the following:

"106 Runout of cartridge assembly NA 3.2 Gage/4.5.5.1"

And add:

"Note 4. To be performed immediately prior to packaging."

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4.5.5: Delete the entire paragraph and replace with the following:

"4.5.5: Profile and alignment check. The profile and alignment of each crimped cartridge shall be verified by a gage corresponding to drawing 8828098, which is to be horizontal ($0^{\circ} \pm 2.5^{\circ}$). The rear face of the cartridge shall be positioned within one inch of the gage's push arm and be inserted with a peak force not to exceed 60 pounds (27.2 kg) and a peak velocity of 2 ft/sec (61 cm/sec). No manual assistance (i.e., rotation, agitation, etc.) is permitted during inspection. Any cartridge which fails to be inserted or seated with the required force and velocity in any orientation and in accordance with the following sampling plan shall be rejected:

a. Inspect (50) cartridges in 4 angular orientations approximately 90° apart.

b. After (50) consecutive cartridges successfully chambergage, switch to chambergaging 1 out of 100 in 4 angular orientations and all others in 1 (one) random orientation.

c. Any failure will result in returning to step (a.)

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When applicable, rejected cartridges which are reworked or repaired in accordance with Government approved procedures shall be reinspected in 4 angular orientations approximately 90° apart. Failure of cartridge to be inserted or seated in any orientation shall be cause for rejection of the cartridge. The profile and alignment gage shall not be utilized as an alignment tool."

* Add the following paragraph:

"4.5.5.1 Runout Inspection. Any cartridge that exceeds the runout dimension per drawing 9343009 shall be chambergaged in 4 angular orientations approximately 90 degrees apart (see 4.5.5). This feature shall be controlled using Statistical Process Controls in accordance with the Contractors Government approval plan."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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